

Partitioning the Projective Plane

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Throughout this post, by projective plane we mean the set of all lines through the origin in $\mathbb{R}^3$.

Definition Say that a subset S of the projective plane is octahedral if all lines in S pass through the closure of two opposite faces of a regular octahedron centered at the origin.

Definition Say that a subset S of the projective plane is weakly octahedral if every set $S' \subseteq S$ such that $|S'|=3$ is octahedral.

Conjecture Suppose that the projective plane can be partitioned into four sets, say S_1, S_2, S_3 and S_4 such that each set S_i is weakly octahedral. Then each S_i is octahedral.

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